

MAMMALS

PHYLUM	Chordata
CLASS	Mammalia
ORDERS	29
FAMILIES	153
SPECIES	5,916

Mammals, the most familiar group of vertebrates, all nourish their young on milk produced by the female's mammary glands (the unique skin structures after which the class is named). Most also give birth to live young and, with only a few exceptions, have a covering of hair on their body. Mammals are most widespread and diverse on land, but they have also colonized the water. Their success is largely due to their ability to maintain a constant internal body temperature, regardless of changing external conditions. Most mammals are also highly adaptable and often modify their behavior to suit changing circumstances. Some mammals, especially primates (the group that includes humans), form complex societies.

Evolution

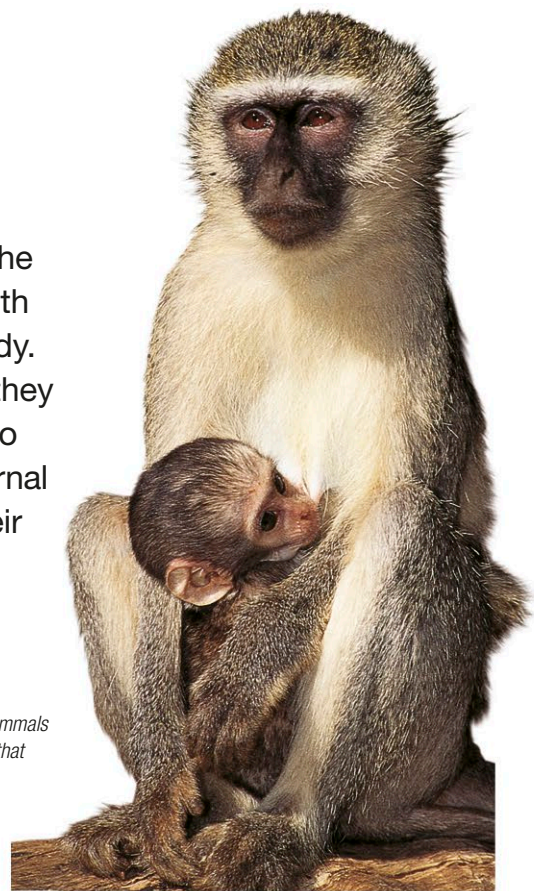
The ancestors of mammals were a group of reptiles known as the therapsids. These small, active carnivores lived during the Triassic Period (which began 251 million years ago). The various features that now separate reptiles and mammals in fact developed over a long period of time and at different rates. In addition to several important differences in skull

structure (see panel, left), the therapsids evolved a lighter, more flexible skeleton. Another significant change was the alignment of the limbs beneath rather than at the sides of the body, which helped the early mammals become faster and more agile.

The transition from reptiles to mammals was completed toward the end of the Triassic Period (195 million years ago), although some early mammals (multituberculates) are said to have started evolving and diversifying 20–40 million years before dinosaurs became extinct. These reptiles were the dominant animals on Earth during the Mesozoic Era (251–65 million years ago), before they were ousted by modern mammals that outcompeted them for food. Mammals' ability to regulate their internal temperature may explain why they outlasted the dinosaurs. During the Mesozoic Era, the climate became cooler and daily temperatures were more variable. Like

YOUNG MAMMALS

One of the key features that sets mammals apart from other animals is the way that females suckle their young on milk from their mammary glands. Some mammals look after their young until they are several years old.



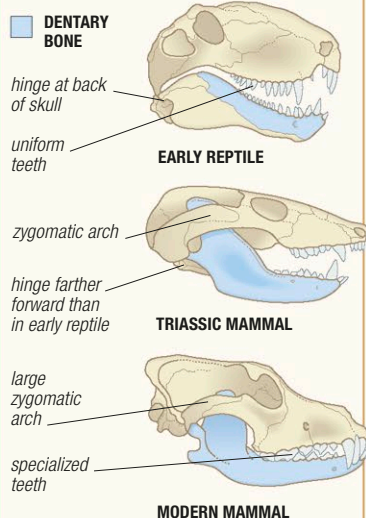
other reptiles, the bodies of most species of dinosaurs would have tended to shut down in low temperatures. Mammals, however, were unaffected by these fluctuations in temperature and so were better able to survive the change in climate.

Anatomy

Several differences between the skeletons of mammals and other vertebrates can be seen in the skull. Mammals are unique in having a lower jaw that is hinged directly to the rest of the skull—in other vertebrates the connection is indirect, with at least one other bone between the 2 parts of the jaw. This direct articulation, and the fact that the lower jaw is made of a single bone (see panel, left), makes the jaw a powerful tool for cutting and dismembering food. Unlike other vertebrates, mammals also have a set of specialized teeth,

which reflects and facilitates their varied diet. All mammals have 3 types of teeth: incisors, which are used for biting; canines for gripping and tearing; and cheek teeth (molars and premolars), which are used for grinding. The shape and size of each type varies according to diet.

Most mammals, unlike other animals, have mobile external ears (pinnae) to locate sounds and then funnel them to the eardrum, where

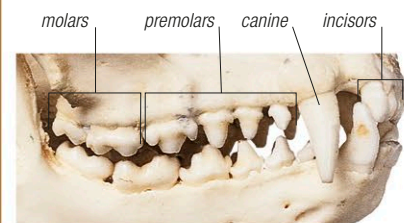
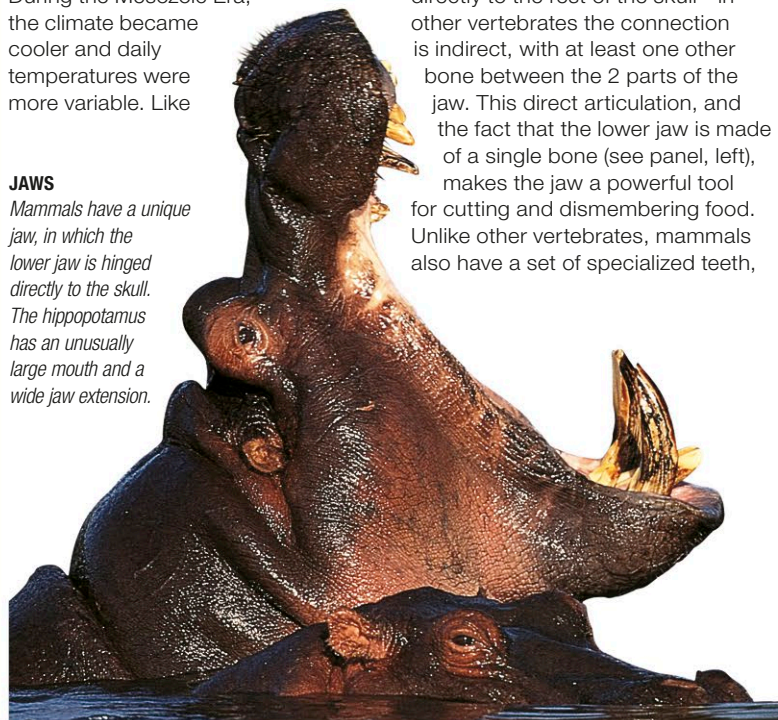


EVOLVING SKULL

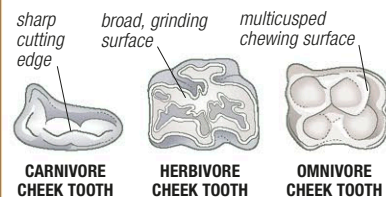
Mammals' reptilian ancestors had uniform teeth and a jaw that was made of several bones and limited to an up-and-down movement. By the Triassic Period, mammal skulls had both a zygomatic arch to house powerful jaw muscles and only one jawbone (the dentary). Modern skulls have specialized teeth and a single jawbone often capable of complex movements. These features, in combination with strong jaw muscles, allow mammals to chew their food—a unique feature of the group.

JAWS

Mammals have a unique jaw, in which the lower jaw is hinged directly to the skull. The hippopotamus has an unusually large mouth and a wide jaw extension.



TEETH OF A RACCOON



TEETH IN MAMMALS

Many mammals, such as raccoons (top), have incisors, canines, and premolars and molars (cheek teeth) that reflect their diet. Carnivores have sharp cheek teeth for cutting bones, while herbivores have broad cheek teeth to grind up vegetation. Omnivore cheek teeth are broad and multicusp for chewing a variety of foods.